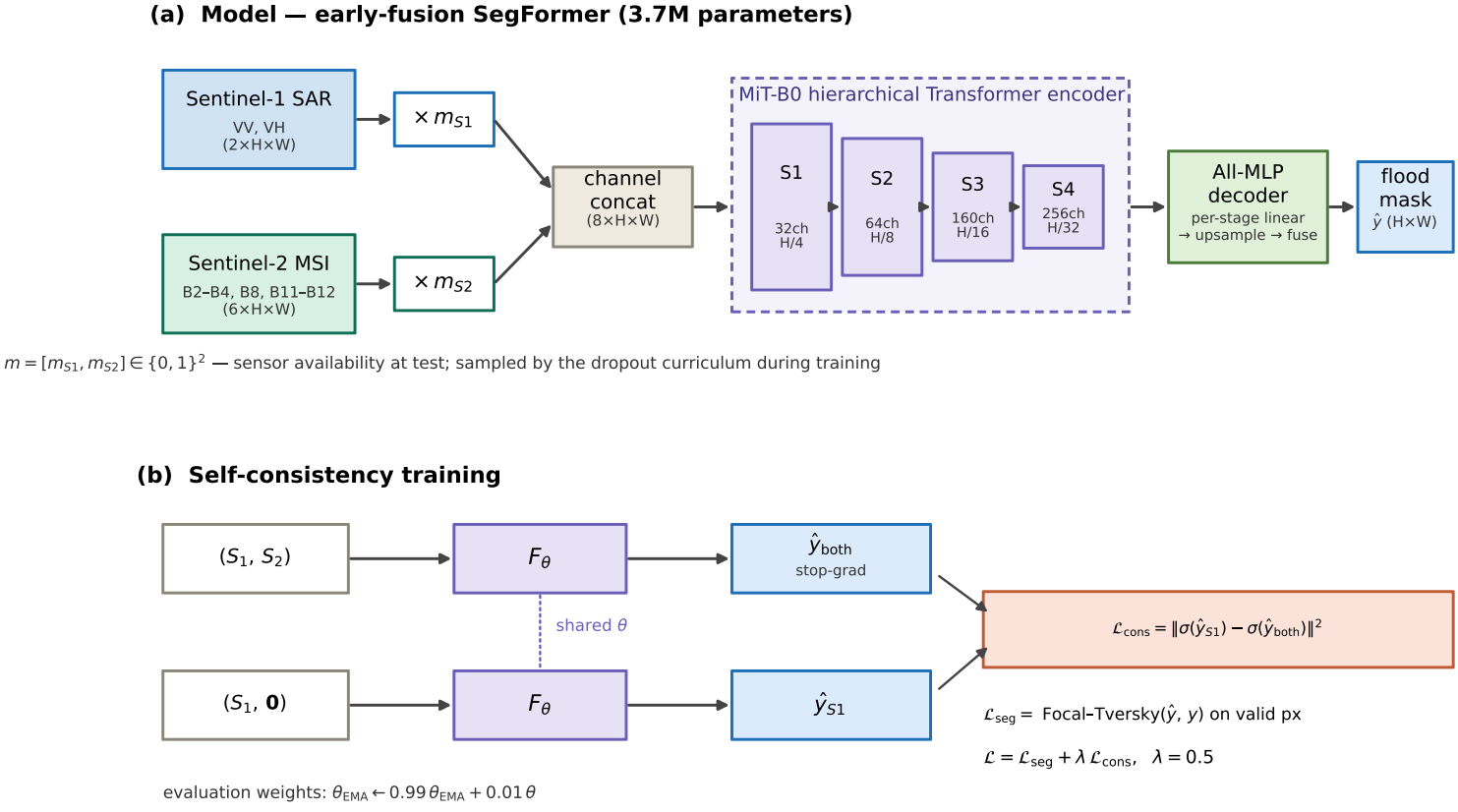
(a) Model — early-fusion SegFormer (3.7M parameters)
Sentinel-1 SAR
VV, VH
(2×H×W)
× m_{S1}
Sentinel-2 MSI
B2–B4, B8, B11–B12
(6×H×W)
× m_{S2}
channel concat
(8×H×W)
MiT-B0 hierarchical Transformer encoder
S1
32ch
H/4
S2
64ch
H/8
S3
160ch
H/16
S4
256ch
H/32
All-MLP decoder
per-stage linear
→ upsample → fuse
flood mask
$\hat{y}$ (H×W)
$m = [m_{S1}, m_{S2}] \in \{0,1\}^2$ — sensor availability at test; sampled by the dropout curriculum during training
(b) Self-consistency training
(S_1, S_2)
F_θ
$\hat{y}_{\text{both}}$
stop-grad
shared θ
$(S_1, \mathbf{0})$
F_θ
$\hat{y}_{S1}$
$\mathcal{L}_{\text{cons}} = \|\sigma(\hat{y}_{S1}) - \sigma(\hat{y}_{\text{both}})\|^2$
$\mathcal{L}_{\text{seg}} =$ Focal–Tversky($\hat{y}$, y) on valid px
$\mathcal{L} = \mathcal{L}_{\text{seg}} + \lambda\,\mathcal{L}_{\text{cons}}, \quad \lambda = 0.5$
evaluation weights: $\theta_{\text{EMA}} \leftarrow 0.99\,\theta_{\text{EMA}} + 0.01\,\theta$